

Multiple Variable Calculus

1. Jacobian

$$\iint dx dy = \iint J dr d\theta$$

J is required for Jacobian (a multiplier) Here, $J = r$

$$\iiint dx dy dz = \iiint r^2 \sin \theta dr d\theta d\phi \quad \text{Jacobian: } r^2 \sin \theta$$

In general, $uv \leftrightarrow \eta, \xi$

Jacobian: A general method for transforming the coordinator of any multiple integral

$$x = x(u, v) \quad y = y(u, v) \quad \text{given}$$

$$\iint_A f(x, y) dx dy$$

$$J(u, v) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$dxdy \rightarrow |J(u, v)| du dv$$

$$J(u, v) = J(x, y; u, v)$$

$$J(u, v) = \left[\frac{\partial(x, y)}{\partial(u, v)} \right]$$

$$u = x + y \quad v = x - y$$

Example: $\iint (x^2 + y^2) dxdy = \iint |J(u, v)| du dv$

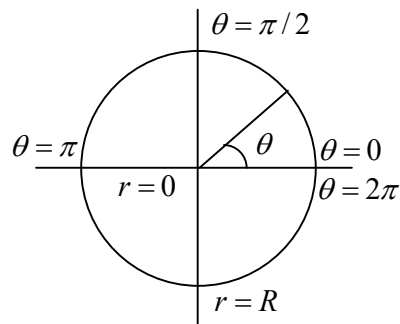
$$x = \frac{u+v}{2}, y = \frac{u-v}{2}; x^2 + y^2 = \left(\frac{u+v}{2}\right)^2 + \left(\frac{u-v}{2}\right)^2$$

$$J(u, v) = \begin{vmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{vmatrix} = \left[\frac{-1}{4} - \frac{1}{4} \right] = \left[\frac{-1}{2} \right]$$

$$J(u, v) = \frac{-1}{2}$$

$$|J(u, v)| = \frac{1}{2}$$

Example: Find area of the circle of radius R



Solution: Let A be the region bounded by a circle of radius R centered at origin. Area of this region is

$$\int_A dA$$

Change into polar coordinates

$$x = r \cos \theta, y = r \sin \theta$$

$$J(r, \theta) = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \cos^2 \theta + r \sin^2 \theta$$

$$|J(r, \theta)| = r$$

$$\int_A dA = \iint |J(r, \theta)| dr d\theta = \int_0^{2\pi} \int_0^R r dr d\theta = \int_0^{2\pi} \left. \frac{r^2}{2} \right|_0^R d\theta = \left. \frac{R^2 \theta}{2} \right|_0^{2\pi}$$

$$\text{Area} = \pi R^2$$

Example: The diamond shaped region 'A' is bounded by the linear

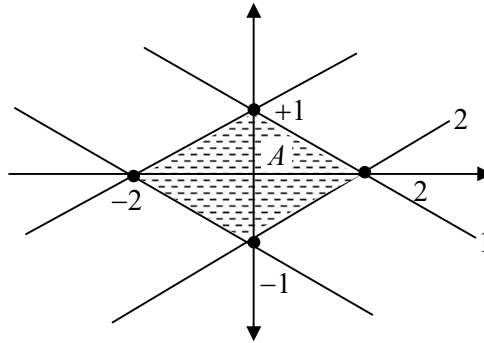
$$x + 2y = 2$$

$$x - 2y = 2$$

$$x + 2y = -2$$

$$x - 2y = -2$$

$$I = \iint (3x + 6y)^2 dx dy$$



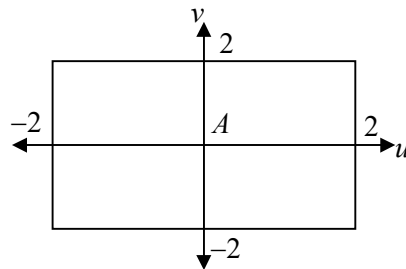
The region is neither vertically nor horizontally symmetrical.

So, we will change the coordinates to make it symmetrical.

$$\text{Transforming} \Rightarrow x + 2y = u \Rightarrow x = \frac{u + v}{2}$$

$$\Rightarrow x - 2y = v \Rightarrow y = \frac{u - v}{4}$$

$$\Rightarrow \begin{cases} u = 2 \\ v = 2 \\ u = -2 \\ v = -2 \end{cases}$$



$$J(u, v) = \begin{vmatrix} 1/2 & 1/2 \\ 1/4 & -1/4 \end{vmatrix} = \frac{-1}{8} - \frac{1}{8} = \frac{-1}{4}$$

$$|J(u, v)| = \frac{1}{4}$$

$$\iint (3x+6y)^2 dx dy = \int_{-2}^2 \int_{-2}^2 \left[\frac{3}{2}(u+v) + \frac{3}{2}(u-v) \right]^2 \frac{1}{4} du dv$$

$$= \int_{-2}^2 \int_{-2}^2 9u^2 \frac{1}{4} du dv = \frac{9}{4} \int_{-2}^2 \frac{u^3}{3} \Big|_{-2}^{+2} dv$$

$$I = \frac{3}{4} \times 16 \times 4 = 48$$

$$I = 48$$

Example: $\iint (x^2 + y^2) dx dy$

(a) $u = x + y, v = x - y$

$$x = \frac{u+v}{2}, y = \frac{u-v}{2}$$

$$x^2 + y^2 = \frac{1}{4} [(u+v)^2 + (u-v)^2] = \frac{1}{4} [2u^2 + 2v^2] = \frac{u^2 + v^2}{2}$$

(b) $J(u, v) = \begin{vmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{vmatrix} = -\frac{1}{2}$

$$|J(u, v)| = \frac{1}{2}$$

(c) $I = \iint \frac{1}{2} (u^2 + v^2) \frac{1}{2} du dv = \iint \frac{1}{4} (u^2 + v^2) du dv$

(d) The limits

(0, 0)	(2, 2)	(-1, 5)	(-3, 3)
$u = 0$	$u = 4$	$u = 4$	$u = 0$
$v = 0$	$v = 0$	$v = -6$	$v = -6$

$$I = \int_{-6}^0 \int_0^4 \frac{1}{4} (u^2 + v^2) du dv = \frac{1}{4} \int_{-6}^0 \left(\frac{u^3}{3} + v^2 u \right) \Big|_0^4 dv$$

$$= \frac{1}{4} \left(\frac{64}{3} v + \frac{4 \cdot v^3}{3} \right) \Big|_{-6}^0 = \frac{1}{4} \left(+\frac{64}{3} \times 6 + \frac{4}{3} \times 6 \times 6 \times 6 \right) = 32 + 72$$

$$I = 104$$

Example: Consider the coordinate transformation

The relation between the area elements $dx'dy' = j dxdy$

$$x' = \frac{x+y}{\sqrt{2}} \quad y' = \frac{x-y}{\sqrt{2}} \quad \text{The value of } |J| = ?$$

Solution: $J(x, y) = \begin{vmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{vmatrix} = \frac{-1}{2} - \frac{1}{2} = -1$

$$|J| = 1 = j$$

$$dxdy = |J| dudv, \quad J(u, v)$$

Example: Consider the transformation to a new set of coordinates (ξ, η)

$$\xi = 2x + 3y$$

$$\eta = 3x - 2y$$

In ξ, η coordinate system, the area element

$$dxdy = ??$$

Solution: $dxdy = |J(\xi, \eta)| d\xi d\eta$

$$J(\xi, \eta) = \begin{vmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} \end{vmatrix}$$

$$J(x, y) = \begin{vmatrix} 2 & 3 \\ 3 & -2 \end{vmatrix}$$

$$J(x, y) = -4 - 9 = -13 \Rightarrow J(\xi, \eta) = \frac{-1}{13}$$

$$\frac{4x+6y}{13x-2\xi+3\eta} = \frac{2\xi+3\eta}{13} = \frac{2\xi+2\eta}{13}$$

$$\frac{3}{13} - \frac{2}{13}$$

$$\frac{-4}{(13)^2} - \frac{9}{(13)^2} = \frac{-1}{13}, \therefore dxdy = \frac{1}{13} d\xi d\eta$$